

Project Report: RESTful Bookstore API

Abstract

This project focuses on developing a RESTful Bookstore API for managing bookstore operations in a secure and organized way. The system is designed to handle books, authors, categories, and user information efficiently. The project uses a layered architecture to separate different functionalities and improve maintainability. DTOs are used to transfer data safely between the database and users. Security is implemented using JWT authentication to provide safe access to resources. Validation and exception handling are also included to ensure stable communication between the system and clients. This project provides a reliable backend service that can easily connect with web or mobile applications.

Introduction

Today, online applications require powerful backend systems to manage large amounts of data and user requests. A bookstore management system requires proper organization of books, users, and related information. The RESTful Bookstore API was developed to create a centralized backend service that follows REST principles and provides secure communication. The main goal of this project is to create a scalable and maintainable system that supports different operations while keeping the code clean and reusable. The project demonstrates practical knowledge of enterprise Java development, database handling, security implementation, and API design.

Tools Used

- Programming Language: Java 17
- Frameworks: Spring Boot 3 (Spring Web, Spring Data JPA, Spring Security)
- Database: H2 In-Memory Database
- Security: JSON Web Tokens (JWT)
- Build Tool: Maven
- Documentation: Swagger / OpenAPI 3.0
- Testing: JUnit 5, Mockito, MockMvc
- Utility Tool: Lombok

Steps Involved in Building the Project

1. Initialized the project using Spring Boot and Maven setup.
2. Configured database properties and JWT settings.
3. Designed entities such as Book, Author, Category, User, and Role.
4. Developed repository classes for database operations.
5. Implemented service classes for handling business logic.
6. Created DTO and Mapper classes for secure data transfer.
7. Developed REST API endpoints for CRUD operations.
8. Added validation for user requests and input checking.
9. Implemented Spring Security with JWT authentication and role-based access.
10. Added global exception handling for proper error responses.
11. Integrated Swagger documentation for API testing.

12. Performed testing using JUnit, Mockito, and MockMvc.

Conclusion

The RESTful Bookstore API successfully meets the requirements of a secure and modern backend system. By using Spring Boot and a layered structure, the application becomes easy to manage and extend. JWT authentication improves security while DTO implementation protects internal database details. Exception handling and testing increase reliability and maintain system stability. This project creates a strong foundation for future improvements such as payment systems, cloud deployment, caching, and frontend integration.